

MAGNETISM

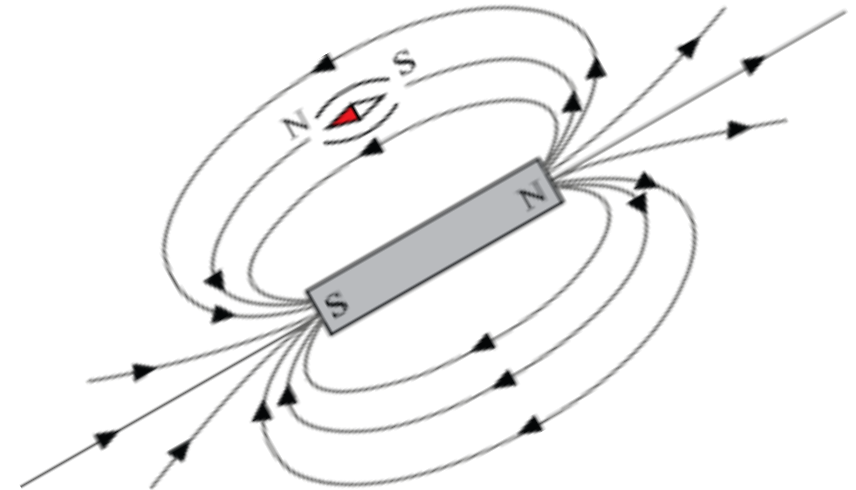
**PHT100 Engineering Physics A
Module IV**

Introduction

- ✓ Magnetism: - Phenomenon of attracting magnetic substances like iron, nickel, cobalt, etc.
- ✓ A body possessing the property of magnetism is called a magnet.
- ✓ A magnetic pole is a point near the end of the magnet where the field lines are concentrated.
- ✓ The region around a magnet in which it exerts forces on other magnets and on objects made of iron is a magnetic field.

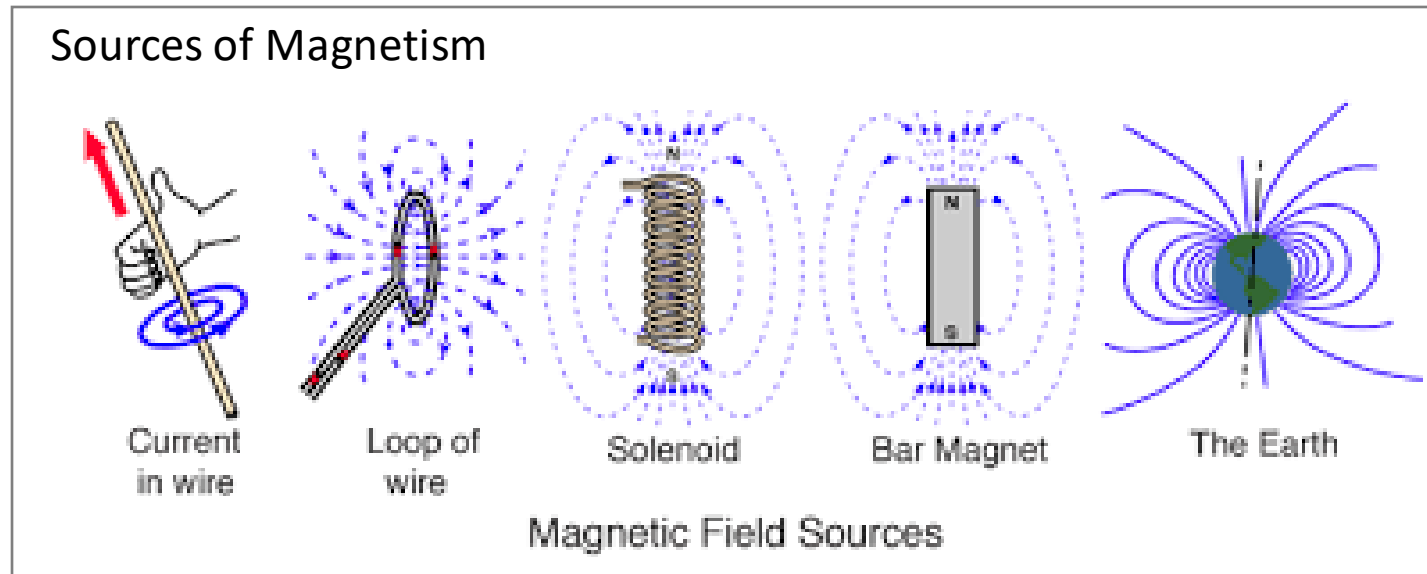
Properties of a Magnet

- A freely suspended magnet aligns itself along North – South direction.
- It attracts magnetic substances.
- Unlike poles attract and like poles repel each other.
- Magnetic poles always exist in pairs. i.e. Poles can not be separated.
- A magnet can induce magnetism in other magnetic substances.



Magnetic Field

- 👉 **It is a vector field in which a moving charge experience a force and magnetic dipole experience a torque.**
- 👉 Magnetic field is represented by the lines of force moving from north pole to south pole of magnet
- 👉 Magnetic field is created by moving charge or magnetized materials



Magnetic Flux

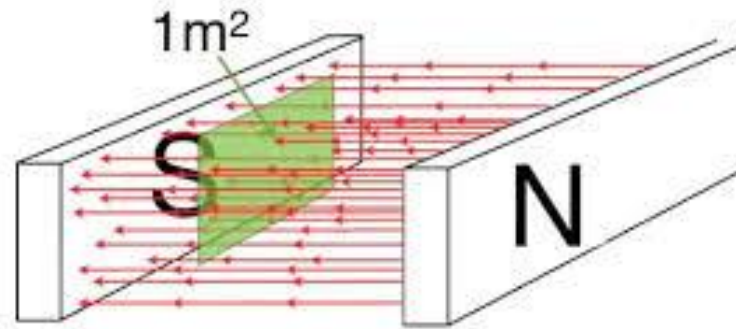
- ❑ Magnetic Flux is defined as the number of magnetic field lines passing through a given closed surface. It gives the measurement of the total magnetic field that passes through a given surface area..
- ❑ It is a scalar quantity & its unit is Weber(Wb)

Magnetic Induction (B)

Magnetic Induction (B) at a point is defined as the magnetic flux passing through unit area around the point. It is also called magnetic flux density.

It is a vector quantity & its unit is Weber/m² or Tesla (T)

$$B = \frac{\phi}{A}$$



Magnetising Field /Magnetic Field Strength (H)

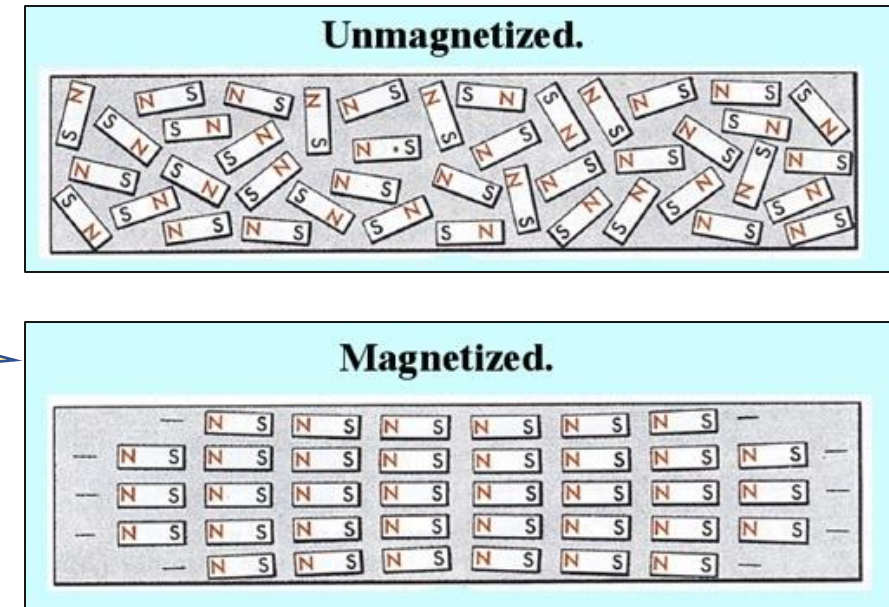
- ✓ It represent magnetic field produced by external current or magnet.
- ✓ It does not include the response of medium to the field.

$$H = \frac{B}{\mu} \quad \text{where } \mu \text{ is the permeability of material}$$

In the presence of magnetizing field (H), material can get magnetized due to alignment of magnetic dipoles

$$B = \mu_0 (H+M)$$

Where μ_0 is the permeability of free space &
M is magnetization.



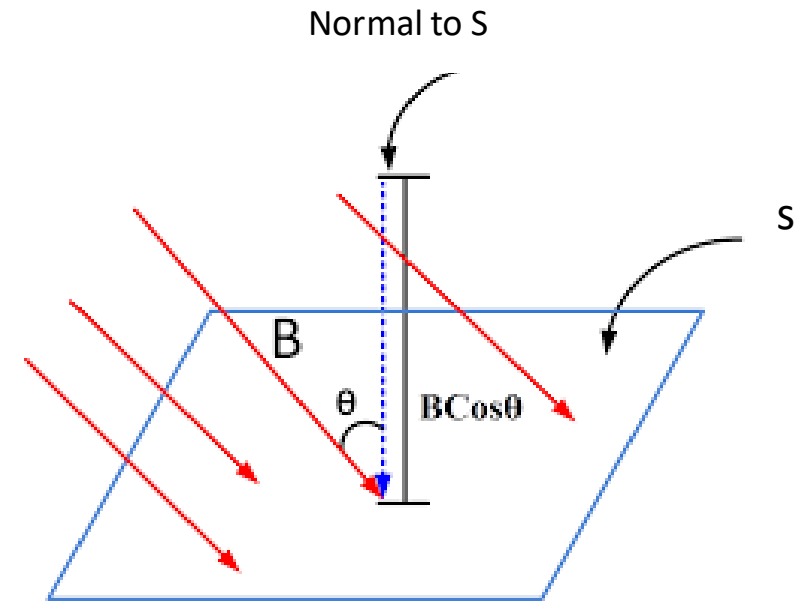
Magnetic Flux

The total magnetic flux through a small elemental area (ds) is

$$d\phi_B = B_{\text{perpendicular}} \cdot ds$$

$$d\phi_B = B \cos \theta \, ds$$

$$d\phi_B = \vec{B} \cdot \vec{ds}$$



Gauss' Law for Magnetic Flux Density

Gauss' Law states that the magnetic field B has divergence equal to zero.

The differential form for Gauss's law for magnetism is:

$$\nabla \cdot B = 0$$

where ∇ denotes divergence, and B is the magnetic field.

The integral form of Gauss's law for magnetism is $\phi_B = \int \vec{B} \cdot \vec{ds} = 0$

This means

- *Number of field lines entering a surface is equal to that leaving it. i.e. magnetic lines of force always form closed loops.*
- *Magnetic monopoles do not exist.*

Ampere's Circuital Law

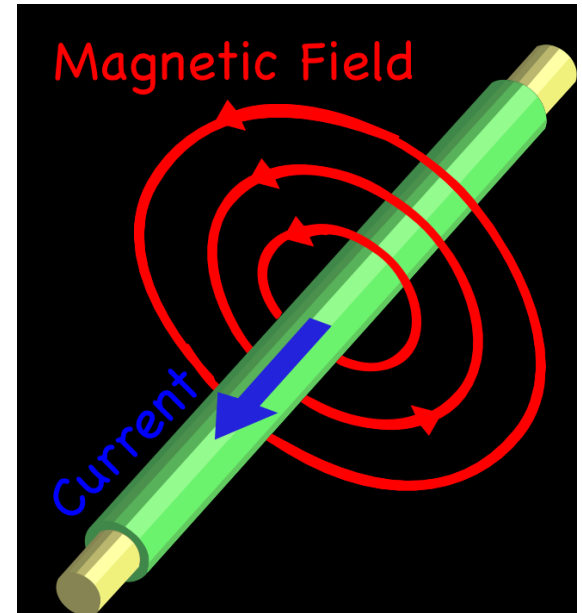
Ampere's circuital law states that line integral of magnetic field forming a closed loop around the current carrying conductor, in the plane normal to the current, is equal to the μ_0 times the net current passing through the closed loop.

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

If there are multiple current carrying conductors in the loop, then algebraic sum of the currents need to be taken into account.

For ex; if there are three current carrying conductors, then

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_1 + I_2 + I_3)$$



Ampere's Circuital Law : Proof

Consider cross section of a current carrying conductor. Current is directed outwards. Magnetic field at a distance r from the conductor is

$$B = \frac{\mu_0}{4\pi} \times \frac{2I}{r}$$

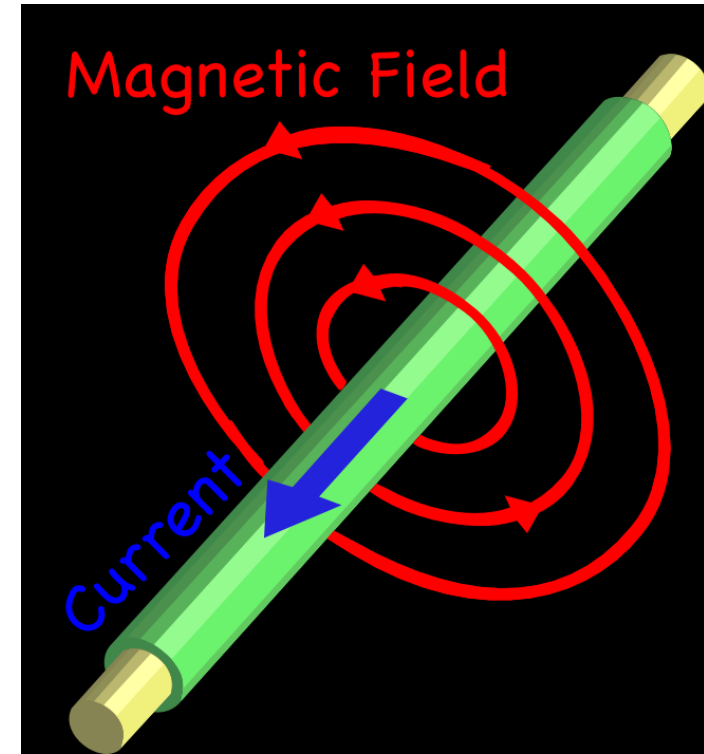
$$B = \frac{\mu_0 I}{2\pi r}$$

Since radius ' r ' is a constant around the Amperian loop, magnetic field (B) will be a constant

$$\oint \mathbf{B} \cdot d\mathbf{l} = \oint B \cos\theta \, dl = \oint B \, dl \quad (\text{Since } B \text{ and } dl \text{ are parallel})$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = B \oint dl = B 2\pi r$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi r} 2\pi r = \mu_0 I$$



Line integral of magnetic field forming a closed loop around the current carrying conductor is

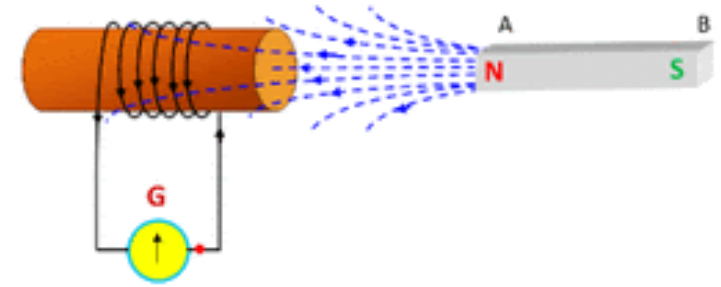
- ☛ dependent only on current
- ☛ independent of the radius of the loop.

Faraday's Law

Faraday's law states that the induced emf is equal to the rate of change of the magnetic flux.

$$\varepsilon = \frac{d\phi_B}{dt}$$

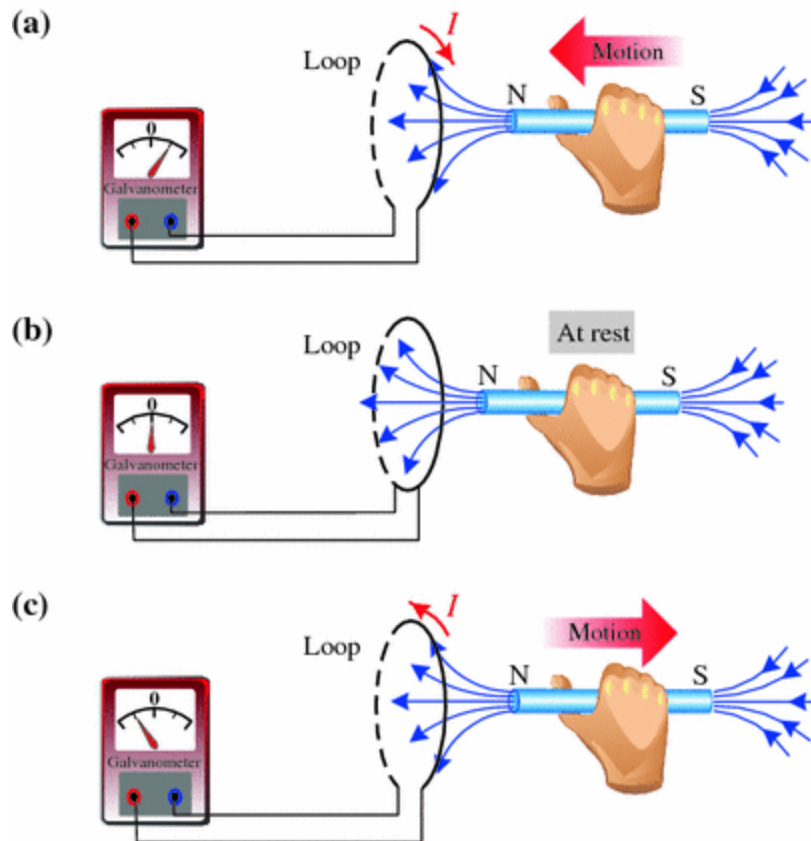
where ε is the electromotive force (EMF) and Φ_B is the magnetic flux.



Lenz's Law

Soon after Faraday proposed his law, Lenz devised a rule—now known as **Lenz's law**, for determining the direction of an induced emf and the direction of an induced current in a loop. This law states that:

An induced current in a loop is created such that the internal magnetic field of the loop opposes the changes in the external magnetic flux.

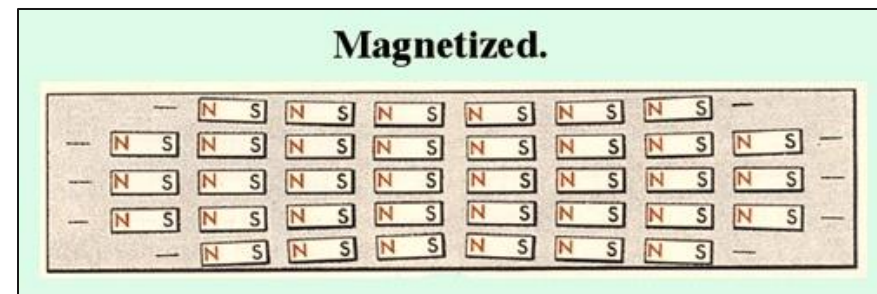
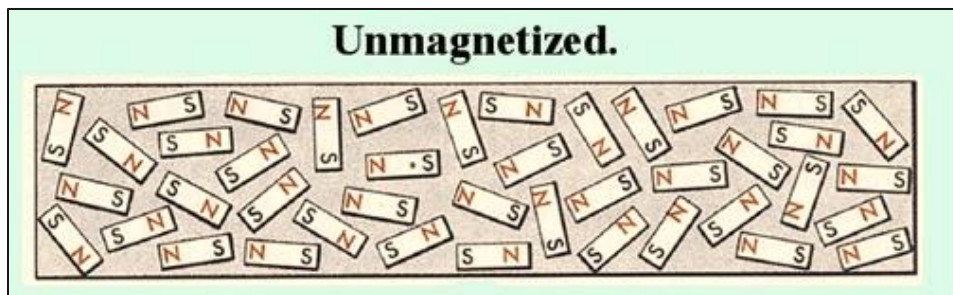


The direction of induced current is such that the magnetic field produced by induced current opposes the change in external magnetic field.

$$\varepsilon = -\frac{d\phi_B}{dt}$$

Magnetisation & Magnetic Susceptibility

Material is magnetized due to rearrangement of atomic or molecular magnetic dipoles.



Magnetisation is the density of permanent or induced magnetic dipoles in a material.

$$\text{Magnetisation} = \frac{\text{Magnetic dipole moments}}{\text{volume of the material}}$$

Magnetic Susceptibility is the degree of magnetisation in response to an applied field.

$$\chi = \frac{M}{H}$$

➡ Magnetic materials are classified as diamagnetic, paramagnetic, or ferromagnetic on the basis of their susceptibilities.

Magnetic Permeability

- ☛ Magnetic permeability (μ) is defined as the property of the material to allow the flow of magnetic lines of force to pass through it.

OR

- ☛ Ratio of magnetic flux density **B** established within the material to the magnetic field strength **H** of the magnetizing field.

$$\mu = \frac{B}{H}$$

- ☛ The SI unit of magnetic permeability is Henry per meter.

Relative Permeability

Relative Permeability is the comparison of permeability of material to that of vacuum or air.

B is the magnetic induction in a material placed in a magnetising field (H)

$$\mu = \frac{B}{H}$$

$$B = \mu H \quad \text{where } \mu \text{ is the permeability of material}$$

B_0 is the magnetic induction at same place when material is removed

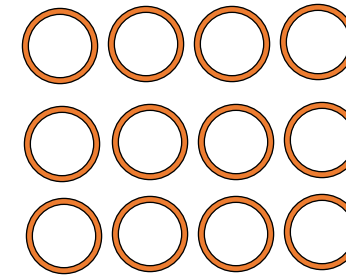
$$B_0 = \mu_0 H \quad \text{where } \mu_0 \text{ is the permeability of free space}$$

☞ Relative permeability is

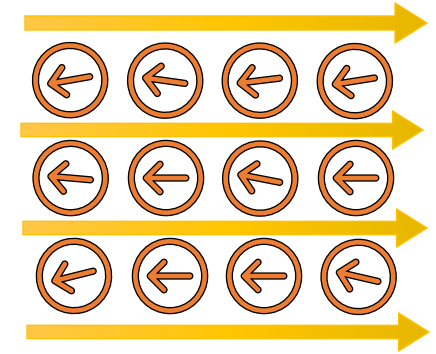
$$\frac{B}{B_0} = \frac{\mu}{\mu_0} = \mu_r$$

Diamagnetism

- Diamagnetic substances are composed of atoms which have no net magnetic moment.
- When exposed to a field, a negative magnetization (dipoles arranged in opposite direction) is produced and thus the susceptibility is negative.



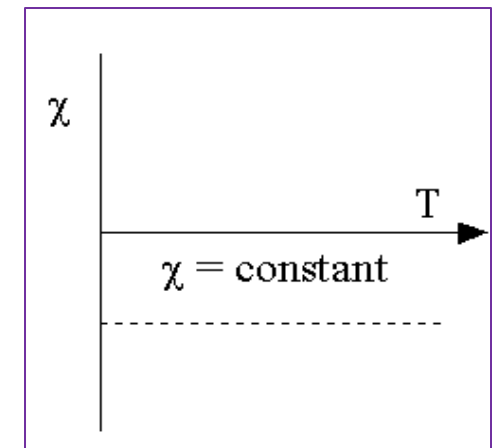
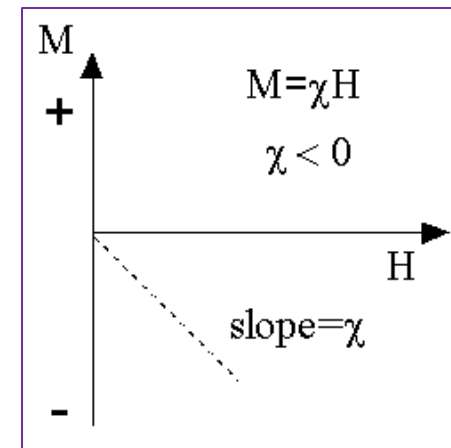
Without external magnetic field



In the presence of magnetic field

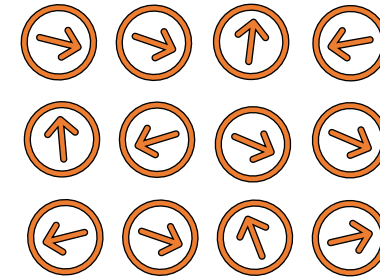
- Susceptibility is negative and is independent of temperature
- They tend to move from region of strong magnetic field to weak field, when placed in a non uniform magnetic field.
- They are weakly repelled by a magnet
- Relative permeability is less than unity.
- Diamagnetism is a fundamental property of all matter & it is usually very weak.

Ex: quartz, calcite, water, bismuth, antimony, gold, silver.

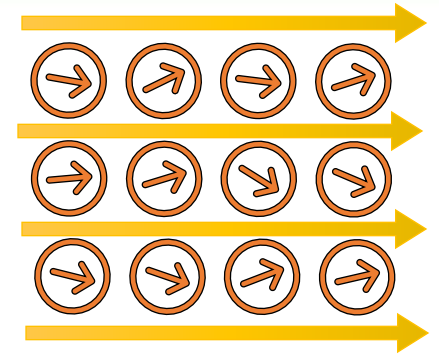


Paramagnetism

- Paramagnetic substances are composed of atoms or molecules which have a net magnetic moment in the absence of external field.
- When exposed to a field, a positive magnetization is produced.



Without external magnetic field



In the presence of magnetic field

- **Susceptibility is small and positive .**
- Susceptibility varies inversely with temperature and substance obeys Curie Law.

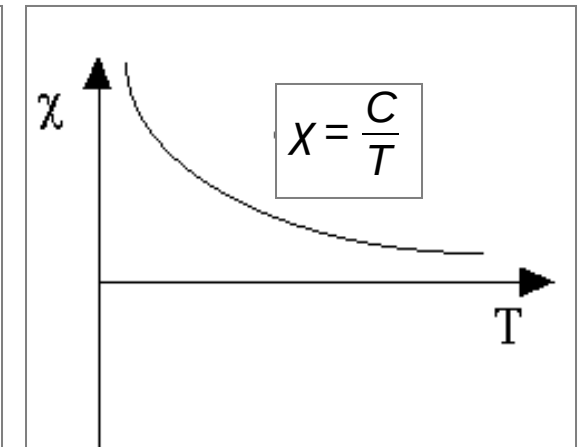
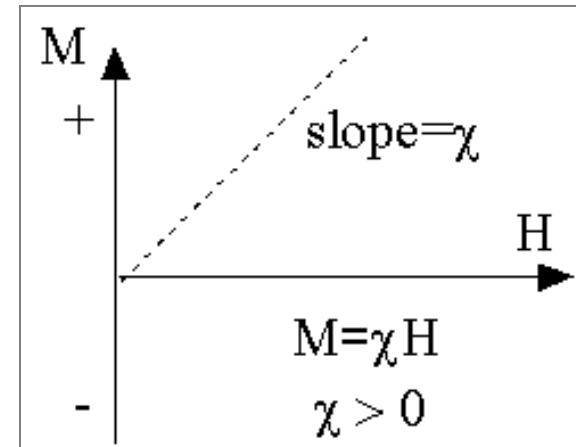
$$M = \frac{CB_{\text{external}}}{T}$$

M is the magnetization

B is the magnetic field, measured in Tesla

T is absolute temperature, measured in kelvin

C is a material-specific Curie constant



- **Relative permeability is greater than unity.**
- They tend to move from region of weak magnetic field to strong field, when placed in a non uniform magnetic field.
- They are weakly attracted by a magnet.

Ex: Lithium, magnesium, oxygen, copper, chromium, platinum.

Ferromagnetism

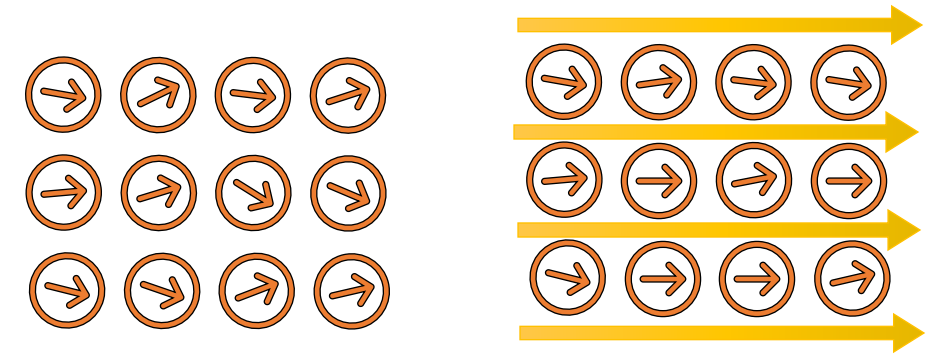
- Ferromagnetic substances are composed of atoms or molecules which have a net magnetic dipole moment in the absence of external field.
- When exposed to a field, magnetic field lines are highly concentrated within the material resulting in a positive magnetization.

- **Susceptibility is large and positive .**
- **Susceptibility varies inversely with temperature**
- All ferromagnetic materials become paramagnetic above a temperature called Curie temperature

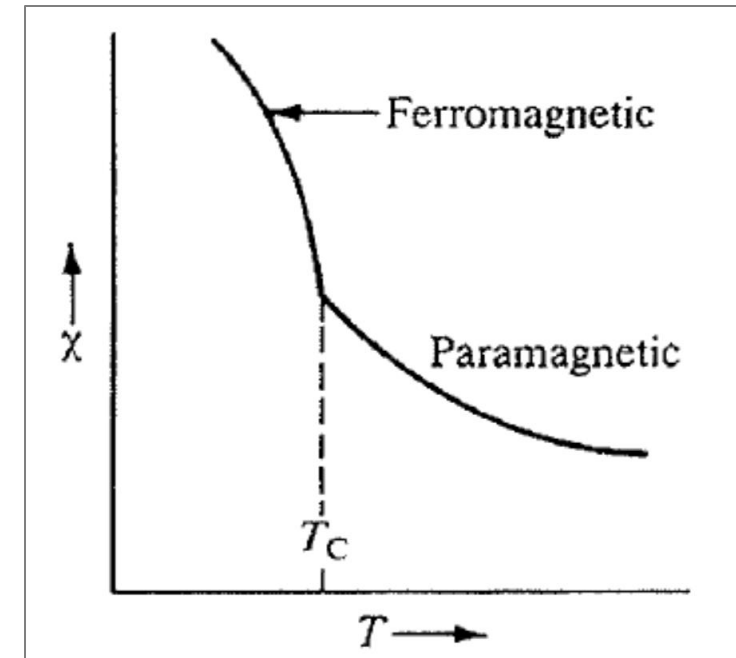
$$\chi = \frac{C}{T - T_c}$$

Where T is absolute temperature, measured in Kelvin
 T_c is Curie temperature
 C is a material-specific Curie constant

- **Relative permeability is greater than unity.**
- **Exhibit the phenomenon of Hysteresis.**
- Ex: Iron , Nickel, Cobalt, Steel , Alnico



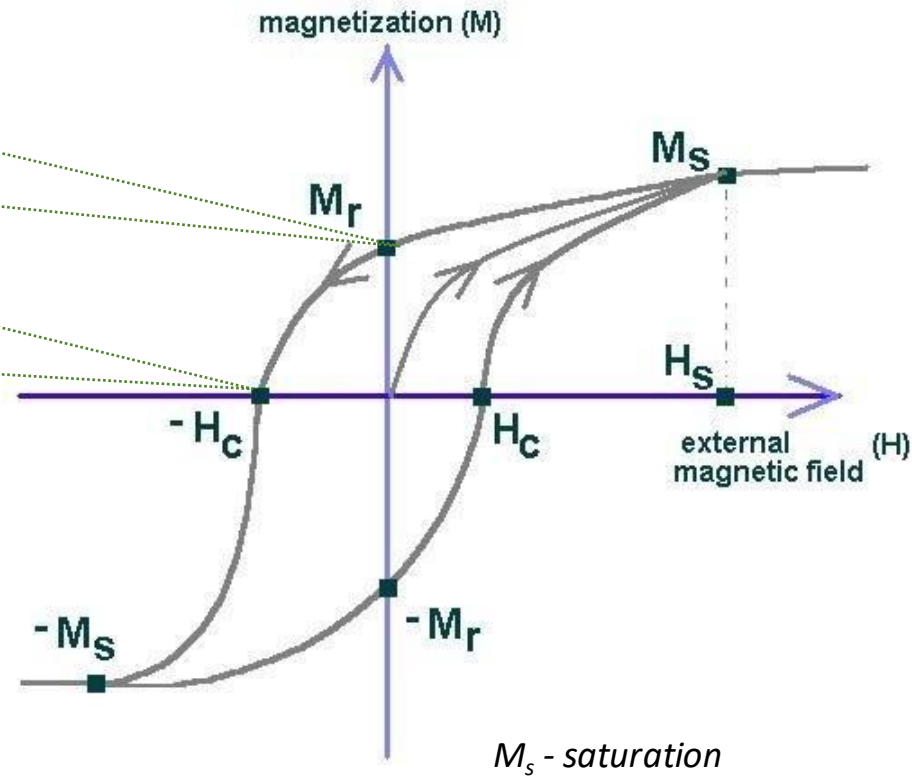
Without external magnetic field In the presence of magnetic field



Hysteresis of Ferromagnetic Material

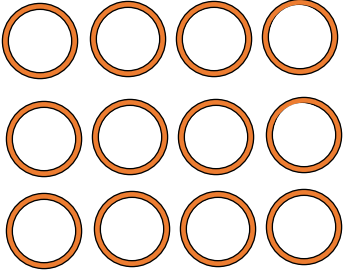
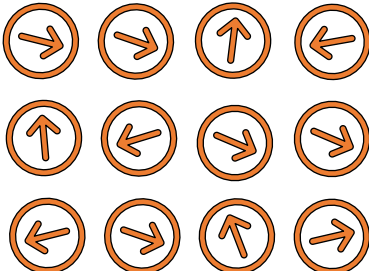
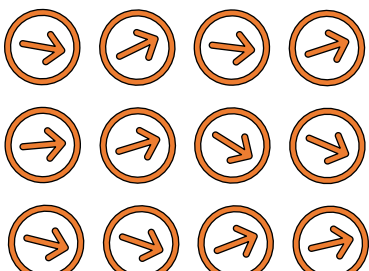
When magnetising field (H) is reduced to zero, the specimen retains some magnetisation (M_r) and this is called retentivity of the material.

Magnetisation becomes zero for certain reverse magnetising field (H_c) This field is called coercivity of the material.



- ❑ When a magnetic field is applied, ferromagnetic material is magnetized in one direction. When this field is withdrawn magnetization do not relax back to zero. This retained magnetisation is known as retentivity of the material. It must be driven back to zero by a field in the opposite direction. This reverse magnetic field is called the coercivity of the material.
- ❑ *If an alternating magnetic field is applied to the material, its magnetization will trace out a loop called a hysteresis loop. The lack of retraceability of the magnetization curve is the property called hysteresis and it is related to the existence of magnetic domains in the material. Once the magnetic domains are reoriented, it takes some energy to turn them back again.*

Dipole moment orientation in Magnetic Materials

Condition	Diamagnetic Material	Paramagnetic Material	Ferromagnetic Material
Without external magnetic field			
In the presence of magnetic field 	